فائل کے تمام سوالات کا مکمل تجزیہ کرنے کے بعد تمام ڈپلیکیٹ اور ناقص (Incomplete/Flawed) سوالات کو خارج کر
کے ایک **فائنل اور شفاف سوالیہ لسٹ** **(Final Cleansed Master List)** نیچے ترتیب دے دی گئی ہے۔
اس لسٹ کو درست ترتیب (Chapter 01 سے Chapter 07) اور ان کے مخصوص **IAE IDs** کے ساتھ فائنل کیا گیا ہے تاکہ
ہر سوال اپنی اصل جگہ پر واضح رہے۔

Chapter 01: Cell Biology & Microscopy

- **IAE11010001:** Magnification and resolution differ in that magnification is the ratio of image size to object size, whereas resolution is the minimum distance between two points to be distinguished as separate.
- **IAE11010002:** Transmission Electron Microscopy (TEM) visualizes internal ultra-structure, whereas Scanning Electron Microscopy (SEM) provides 3D surface topographies.
- **IAE11010003:** Cell fractionation isolates organelles by differential centrifugation based on density and size.
- **IAE11010004:** The fluid mosaic model describes the cell membrane as a phospholipid bilayer with embedded dynamic proteins.
- **IAE11010005:** Fluidity of animal cell membranes is modulated by cholesterol levels and fatty acid saturation.
- **IAE11010006:** Simple diffusion requires no energy, whereas active transport moves solutes against a concentration gradient using ATP.
- **IAE11010007:** Primary active transport uses ATP directly, while secondary active transport relies on ionic gradients generated by primary transport.
- **IAE11010008:** Facilitated diffusion relies on transmembrane proteins to transport polar molecules down their concentration gradient.
- **IAE11010009:** Endocytosis internalizes bulk materials via vesicle formation, whereas exocytosis secretes cellular contents via vesicle fusion.
- **IAE11010010:** Phagocytosis engulfs large particulate matter, whereas pinocytosis internalizes extracellular fluid containing dissolved solutes.
- **IAE11010011:** Plant cell walls consist primarily of cellulose fibers embedded in a matrix of hemicellulose and pectin.
- **IAE11010012:** Plasmodesmata are cytoplasmic channels that penetrate plant cell walls to enable intercellular transport.
- **IAE11010013:** The nucleus is bounded by a double-membrane envelope containing pores that regulate macromolecular traffic.
- **IAE11010014:** The nucleolus is the sub-nuclear site responsible for ribosomal RNA (rRNA) synthesis and ribosome assembly.
- **IAE11010015:** Rough Endoplasmic Reticulum (RER) synthesizes membrane-bound proteins, whereas Smooth Endoplasmic Reticulum (SER) synthesizes lipids and detoxifies xenobiotics.
- **IAE11010016:** The Golgi apparatus modifies, sorts, and packages proteins received from the RER for transport.
- **IAE11010017:** Lysosomes contain hydrolytic enzymes active at acidic pH for intracellular digestion.
- **IAE11010018:** Peroxisomes perform metabolic oxidations producing hydrogen peroxide, which is subsequently converted to water by catalase.
- **IAE11010019:** Plant vacuoles maintain turgor pressure and store hydrolytic enzymes and metabolites.
- **IAE11010020:** Mitochondria contain inner membrane folds termed cristae that house the

respiratory chain and ATP synthase.
- **IAE11010021:** Chloroplasts feature thylakoid membrane systems organized into grana within a liquid stroma.
- **IAE11010022:** The endosymbiotic theory posits that mitochondria and chloroplasts evolved from engulfed prokaryotes.
- **IAE11010023:** Microtubules (tubulin) direct organelle movement, microfilaments (actin) facilitate cell motility, and intermediate filaments maintain structural integrity.
- **IAE11010024:** Centrosomes act as microtubule-organizing centers containing twin, perpendicularly arranged centrioles in animal cells.
- **IAE11010025:** Cilia are short, numerous projections that move fluid across surfaces, whereas flagella are long, sparse structures driving cell locomotion.
- **IAE11010026:** Tight junctions seal intercellular spaces, desmosomes provide mechanical anchoring, and gap junctions allow direct chemical communication.
- **IAE11010027:** Apoptosis is programmed cell death involving regulated execution, whereas necrosis is unprogrammed lysis driven by extrinsic injury.
- **IAE11010028:** Stem cells possess unlimited self-renewal capacity and differentiation potential, whereas somatic cells are terminally differentiated.

Chapter 02: Biochemistry & Biomolecules

- **IAE11010029:** Water exhibits high specific heat capacity due to an extensive intermolecular hydrogen-bonding network.
- **IAE11010030:** Water acts as a universal solvent for hydrophilic substances by forming hydration shells around polar molecules and ions.
- **IAE11010031:** Monosaccharides are classified based on carbonyl position (aldose vs. ketose) and carbon chain length.
- **IAE11010032:** Alpha-glucose forms helical starch polymers, whereas beta-glucose forms straight-chain cellulose polymers linked by beta-1,4-glycosidic bonds.
- **IAE11010033:** Disaccharides are synthesized by condensation reactions yielding glycosidic bonds and releasing water.
- **IAE11010034:** Starch functions as energy storage in plants, glycogen in animals, and cellulose provides plant structural support.
- **IAE11010035:** Triglycerides consist of a glycerol backbone esterified to three fatty acid chains.
- **IAE11010036:** Saturated fatty acids lack carbon-carbon double bonds, whereas unsaturated fatty acids contain one or more double bonds.
- **IAE11010037:** Phospholipids possess an amphipathic nature with a hydrophilic phosphate head and hydrophobic hydrocarbon tails.
- **IAE11010038:** Steroids feature a core structure of four fused carbon rings.
- **IAE11010039:** Amino acids consist of a central alpha-carbon bound to an amino group, a carboxyl group, a hydrogen atom, and a variable R group.
- **IAE11010040:** Peptide bonds are formed via condensation between the carboxyl group of one amino acid and the amino group of another.
- **IAE11010041:** Primary protein structure is defined by the linear amino acid sequence held by covalent peptide bonds.
- **IAE11010042:** Secondary protein structure includes alpha-helices and beta-pleated sheets stabilized by backbone hydrogen bonding.
- **IAE11010043:** Tertiary protein structure represents 3D folding driven by hydrophobic

interactions, ionic bonds, disulfide bridges, and hydrogen bonds.

- **IAE11010044:** Quaternary structure consists of the functional assembly of multiple polypeptide subunits.
- **IAE11010045:** Protein denaturation involves structural unfolding and loss of native biological activity without breaking primary peptide bonds.
- **IAE11010046:** Nucleotides are composed of a nitrogenous base, a pentose sugar, and one to three phosphate groups.
- **IAE11010047:** DNA contains deoxyribose and thymine in a double helix, whereas RNA contains ribose and uracil usually as single strands.
- **IAE11010048:** Phosphodiester bonds link the 3' hydroxyl of one nucleotide sugar to the 5' phosphate of another.
- **IAE11010049:** ATP releases free energy upon hydrolysis of its terminal phosphoanhydride bond.
- **IAE11010050:** Messenger RNA carries genetic code, transfer RNA transports amino acids, and ribosomal RNA forms ribosomal structural cores.
- **IAE11010051:** Conjugated molecules (e.g., glycoproteins, glycolipids, lipoproteins) combine carbohydrate or lipid moieties with proteins.
- **IAE11010052:** Buffers maintain stable pH by absorbing or releasing free hydrogen ions.
- **IAE11010053:** Hydrolysis reactions break chemical bonds via water addition, whereas condensation reactions synthesize bonds releasing water.

Chapter 03: Enzymes

- **IAE11010054:** Enzymes act as biological catalysts by lowering the activation energy barrier of chemical reactions.
- **IAE11010055:** The active site provides a specific chemical environment tailored to bind specific substrate molecules.
- **IAE11010056:** The lock-and-key model proposes a rigid active site, whereas the induced-fit model describes dynamic structural adaptation upon substrate binding.
- **IAE11010057:** Enzymes increase reaction rates by stabilizing transition states without altering the overall free energy change (\Delta G).
- **IAE11010058:** Enzyme activity increases with temperature up to an optimum, beyond which thermal denaturation occurs.
- **IAE11010059:** Extreme pH alters active site ionization states and breaks ionic bonds, leading to loss of catalytic activity.
- **IAE11010060:** Substrate concentration increases reaction rate linearly until enzyme active sites reach full saturation (V_{max}).
- **IAE11010061:** Competitive inhibitors bind to the active site directly, whereas non-competitive inhibitors bind to an allosteric site.
- **IAE11010062:** Competitive inhibition can be overcome by increasing substrate concentration, whereas non-competitive inhibition cannot.
- **IAE11010063:** Allosteric regulation involves effector molecules binding outside active sites to induce conformational changes.
- **IAE11010064:** Feedback inhibition occurs when an end-product allosterically inhibits an enzyme operating early in its biosynthetic pathway.
- **IAE11010065:** Coenzymes are organic non-protein helpers, whereas cofactors include inorganic metal ions required for enzyme activity.
- **IAE11010066:** Prosthetic groups are tightly or covalently bound non-protein cofactors

essential to enzyme function.

- **IAE11010067:** An apoenzyme is catalytically inactive without its bound cofactor, whereas a holoenzyme is the complete active enzyme-cofactor complex.
- **IAE11010068:** The Michaelis constant (K_m) represents the substrate concentration at which reaction velocity reaches half of V_{max}.
- **IAE11010069:** High K_m indicates low enzyme-substrate affinity, whereas low K_m reflects strong affinity.
- **IAE11010070:** Zymogens are inactive enzyme precursors requiring specific proteolytic cleavage for functional activation.
- **IAE11010071:** Ribozymes are catalytic RNA molecules capable of accelerating specific chemical reactions.
- **IAE11010072:** Isozymes are structurally distinct forms of an enzyme that catalyze the same biochemical reaction.
- **IAE11010073:** Reversible inhibitors bind via non-covalent interactions, whereas irreversible inhibitors form covalent bonds inactivating enzymes permanently.
- **IAE11010074:** Substrate binding can induce strain in substrate bonds, facilitating transition state attainment.
- **IAE11010075:** Heavy metal ions cause non-competitive denaturation by disrupting disulfide linkages and ionic bonds.
- **IAE11010076:** Enzyme specificity is determined by complementary shape, charge, and hydrophobicity of the active site cavity.
- **IAE11010077:** Compartmentalization isolates enzymes within specific organelles to regulate metabolic pathways.
- **IAE11010078:** Industrial enzymes are immobilized on solid matrices to enhance stability and facilitate product separation.
- **IAE11010079:** Product concentration buildup can slow reaction rates via product inhibition mechanism.

Chapter 04: Bioenergetics

- **IAE11010080:** Photosynthesis converts light energy to chemical energy, while respiration oxidizes organic molecules to extract usable ATP.
- **IAE11010081:** Chlorophylls absorb blue and red light while reflecting green wavelengths.
- **IAE11010082:** Accessory pigments expand the action spectrum of photosynthesis by transferring captured energy to reaction centers.
- **IAE11010083:** Photosystem II (P680) photolyzes water, whereas Photosystem I (P700) reduces $NADP^+$ to NADPH.
- **IAE11010084:** Non-cyclic photophosphorylation produces ATP, NADPH, and O_2, whereas cyclic photophosphorylation generates ATP exclusively.
- **IAE11010085:** Photolysis of water releases protons, electrons, and oxygen gas inside the thylakoid lumen.
- **IAE11010086:** Chemiosmosis generates ATP via proton motive force driving ATP synthase across thylakoid/inner mitochondrial membranes.
- **IAE11010087:** The Calvin cycle fixes CO_2 using RuBisCO into 3-phosphoglycerate within the chloroplast stroma.
- **IAE11010088:** The Calvin cycle consumes ATP and NADPH to reduce 3-PGA to glyceraldehyde-3-phosphate (G3P).
- **IAE11010089:** RuBisCO catalyzes primary carbon fixation but exhibits oxygenase activity

leading to photorespiration.

- **IAE11010090:** C_4 plants spatially segregate initial carbon fixation (mesophyll) from the Calvin cycle (bundle sheath) using PEP carboxylase.
- **IAE11010091:** CAM plants temporally separate carbon fixation (night) from Calvin cycle activity (day) to minimize transpiration.
- **IAE11010092:** Glycolysis oxidizes glucose to two pyruvate molecules in the cytosol, yielding net 2 ATP and 2 NADH.
- **IAE11010093:** Pyruvate oxidation converts pyruvate into acetyl-CoA, producing CO_2 and NADH in the mitochondrial matrix.
- **IAE11010094:** The Krebs cycle oxidizes acetyl-CoA, yielding CO_2, NADH, FADH_2, and ATP/GTP per cycle.
- **IAE11010095:** The electron transport chain passes electrons through membrane complexes, establishing a transmembrane proton gradient.
- **IAE11010096:** Oxygen acts as the terminal electron acceptor in aerobic respiration, reducing to form water.
- **IAE11010097:** Lactic acid fermentation reduces pyruvate to lactate, regenerating NAD^+ under anaerobic conditions.
- **IAE11010098:** Alcoholic fermentation converts pyruvate to ethanol and CO_2 to sustain anaerobic glycolysis.
- **IAE11010099:** Aerobic respiration yields up to 32 ATP per glucose, compared to a net yield of 2 ATP in fermentation.
- **IAE110100100:** Substrate-level phosphorylation transfers phosphate directly from a substrate to ADP, unlike oxidative phosphorylation.
- **IAE110100101:** Fatty acids enter aerobic pathways via beta-oxidation, breaking down into acetyl-CoA units.
- **IAE110100102:** Amino acids undergo deamination before their carbon skeletons enter glycolysis or the Krebs cycle.
- **IAE110100103:** Phosphofructokinase acts as a key allosteric rate-limiting enzyme in glycolysis, inhibited by ATP and citrate.
- **IAE110100104:** The light absorption spectrum measures light absorbed vs. wavelength, whereas the action spectrum measures photosynthetic rate vs. wavelength.
- **IAE110100105:** Uncoupling proteins dissipate the mitochondrial proton gradient as heat rather than synthesizing ATP.
- **IAE110100106:** Anaerobic respiration utilizes alternative terminal electron acceptors like nitrate or sulfate instead of oxygen.
- **IAE110100107:** Photorespiration consumes ATP and organic carbon without producing sugars, decreasing photosynthetic efficiency.
- **IAE110100108:** Electron carriers NADH and FADH_2 yield approximately 2.5 and 1.5 ATP respectively via oxidative phosphorylation.
- **IAE110100109:** Cyanide inhibits cytochrome c oxidase, blocking the electron transport chain completely.
- **IAE110100110:** The compensation point occurs when photosynthetic carbon fixation equals respiratory carbon release.
- **IAE110100111:** High CO_2 concentrations suppress photorespiration by favoring RuBisCO carboxylase activity.
- **IAE110100112:** Glyceraldehyde-3-phosphate (G3P) acts as the direct precursor for glucose synthesis in photosynthesis.
- **IAE110100113:** Cyclic electron flow protects photosystem structures against

photoinhibition under stress conditions.
- **IAE110100114:** Matrix pH is higher than intermembrane space pH during active mitochondrial respiration.
- **IAE110100115:** Thermogenin in brown adipose tissue relies on uncoupled respiration for non-shivering thermogenesis.
- **IAE110100116:** Glycerol enters glycolysis at the dihydroxyacetone phosphate (DHAP) intermediate step.
- **IAE110100117:** High ATP/ADP ratios allosterically inhibit key respiratory enzymes to conserve energy reserves.
- **IAE110100118:** Malate functions as the CO_2 storage vehicle in C_4 and CAM metabolic pathways.
- **IAE110100119:** Thylakoid membranes maintain proton accumulation strictly within the internal lumen.

Chapter 05: Acellular Life

- **IAE110100120:** Viruses are obligate intracellular entities consisting of genetic material enclosed in a protein capsid.
- **IAE110100121:** Viral genomes may consist of single- or double-stranded DNA or RNA.
- **IAE110100122:** Viral capsids are composed of protein subunits called capsomeres arranged in helical or icosahedral symmetry.
- **IAE110100123:** Viral envelopes are derived from host cell membranes during budding and contain viral glycoproteins.
- **IAE110100124:** Bacteriophages are viruses that specifically infect bacterial hosts.
- **IAE110100125:** The lytic cycle culminates in host cell destruction, whereas the lysogenic cycle integrates viral DNA into the host chromosome as a prophage.
- **IAE110100126:** Reverse transcriptase synthesizes complementary DNA (cDNA) from an RNA template in retroviruses.
- **IAE110100127:** HIV targets helper T lymphocytes by binding specifically to CD4 receptors and chemokine co-receptors.
- **IAE110100128:** Prions are infectious misfolded proteins that induce conformational conversion of normal cellular proteins.
- **IAE110100129:** Viroids are infectious agent particles consisting solely of unencapsulated circular single-stranded RNA.
- **IAE110100130:** Retroviral integrase inserts reverse-transcribed viral cDNA directly into the host genome.
- **IAE110100131:** Antigenic drift involves minor point mutations, whereas antigenic shift represents major genomic reassortment.
- **IAE110100132:** Temperate phages are capable of undergoing both lytic and lysogenic reproductive cycles.
- **IAE110100133:** Viroids lack capsids and protein-encoding capacity, acting via RNA interference mechanism.
- **IAE110100134:** Prion diseases cause transmissible spongiform encephalopathies without provoking host immune responses.
- **IAE110100135:** Viral tropism is dictated by specific host cell surface receptor availability.
- **IAE110100136:** Latent viral infections remain dormant inside host cells without active viral particle production.
- **IAE110100137:** Plant viruses enter host tissues predominantly through physical wounds

or vector transmission.

- **IAE110100138:** Neuraminidase facilitates the release of progeny influenza viruses from host cell surface membranes.
- **IAE110100139:** Satellite viruses require helper viruses to provide essential functions for replication.

Chapter 06: Prokaryotes

- **IAE110100140:** Prokaryotic cells lack a membrane-bound nucleus and membrane-bound organellar compartments.
- **IAE110100141:** Bacterial cell walls contain peptidoglycan composed of alternating NAG and NAM cross-linked by amino acids.
- **IAE110100142:** Gram-positive bacteria feature thick peptidoglycan layers, whereas Gram-negative bacteria have thin peptidoglycan surrounded by an outer lipopolysaccharide membrane.
- **IAE110100143:** Gram-negative outer membranes contain endotoxins (Lipid A) that provoke inflammatory responses.
- **IAE110100144:** Bacterial capsules prevent phagocytosis by host immune cells.
- **IAE110100145:** Pili mediate genetic exchange via conjugation, whereas fimbriae assist surface attachment.
- **IAE110100146:** Bacterial flagella rotate via a proton-driven basal motor to enable taxis movement.
- **IAE110100147:** Flagellar arrangements are classified as monotrichous, lophotrichous, amphitrichous, or peritrichous.
- **IAE110100148:** Prokaryotic ribosomes are 70S (50S + 30S subunits), whereas eukaryotic ribosomes are 80S (60S + 40S).
- **IAE110100149:** Endospores are highly resistant dormant structures formed by certain bacteria under environmental stress.
- **IAE110100150:** Binary fission is an asexual division process yielding two genetically identical daughter prokaryotes.
- **IAE110100151:** Transformation involves uptake of naked foreign DNA from the surrounding environment.
- **IAE110100152:** Transduction is the transfer of bacterial genes between cells mediated by bacteriophages.
- **IAE110100153:** Conjugation transfers plasmid DNA between bacteria via direct contact through a sex pilus.
- **IAE110100154:** Plasmids are small, extrachromosomal, self-replicating circular DNA molecules encoding non-essential advantageous genes.
- **IAE110100155:** Cyanobacteria perform oxygenic photosynthesis using phycobiliproteins and thylakoid membranes.
- **IAE110100156:** Nitrogen-fixing bacteria convert atmospheric nitrogen gas into bioavailable ammonia using nitrogenase.
- **IAE110100157:** Archaea possess ether-linked membrane lipids and lack peptidoglycan cell walls.
- **IAE110100158:** Methanogens produce methane as a metabolic byproduct under strict anaerobic conditions.
- **IAE110100159:** Halophiles adapt to extreme salinity using specialized ion pumps and compatible solutes.

- **IAE110100160:** Thermophiles stabilize proteins and nucleic acids using heat-stable enzymes and specialized chaperones.
- **IAE110100161:** Nitrifying bacteria oxidize ammonia to nitrite and nitrate to gain chemical energy.
- **IAE110100162:** Penicillin inhibits peptidoglycan cross-linking enzymes in growing bacterial cell walls.
- **IAE110100163:** Streptomycin targets the 30S ribosomal subunit, causing translation errors and inhibition.
- **IAE110100164:** Bacterial growth curves display lag, log (exponential), stationary, and decline phases.
- **IAE110100165:** Obligate aerobes require oxygen, obligate anaerobes are poisoned by oxygen, and facultative anaerobes adapt to both conditions.
- **IAE110100166:** Biofilms are structured microbial communities enclosed within a self-produced extracellular polymeric matrix.
- **IAE110100167:** Quorum sensing regulates bacterial gene expression based on cell population density via signal molecules.
- **IAE110100168:** Mycoplasmas are bacteria that naturally lack cell walls and exhibit pleomorphism.
- **IAE110100169:** Magnetotactic bacteria contain magnetosomes to navigate along geomagnetic field lines.
- **IAE110100170:** Chemolithotrophs obtain energy by oxidizing inorganic compounds such as hydrogen sulfide or iron.
- **IAE110100171:** Heterocysts are specialized cells in cyanobacteria dedicated to nitrogen fixation under aerobic conditions.
- **IAE110100172:** Acid-fast staining identifies bacteria with cell walls rich in mycolic acids (e.g., Mycobacterium).
- **IAE110100173:** R-plasmids carry genes conferring resistance to specific antimicrobial agents.
- **IAE110100174:** Plasmolysis occurs when bacterial cells lose water in hypertonic environments, inhibiting growth.
- **IAE110100175:** CRISPR-Cas systems provide adaptive immunity in prokaryotes against phage infections.
- **IAE110100176:** Hfr strains integrate the F plasmid into their main bacterial chromosome, mediating high-frequency recombination.
- **IAE110100177:** Endotoxin release occurs upon cell lysis of Gram-negative bacteria.
- **IAE110100178:** Microaerophiles require low concentrations of oxygen for growth.
- **IAE110100179:** Bacterial chromosomes consist of single, circular double-stranded DNA localized in the nucleoid region.
- **IAE110100180:** Generalized transduction transfers random bacterial DNA fragments, whereas specialized transduction transfers specific genomic loci adjacent to prophages.
- **IAE110100181:** Teichoic acids present in Gram-positive cell walls maintain structural integrity and charge balance.
- **IAE110100182:** Exotoxins are secreted toxic proteins produced by living Gram-positive or Gram-negative cells.
- **IAE110100183:** Broad-spectrum antibiotics affect both Gram-positive and Gram-negative organisms, whereas narrow-spectrum agents target specific groups.
- **IAE110100184:** Mesosomes are plasma membrane invaginations involved in septum formation and DNA segregation.

- **IAE110100185:** Anoxygenic photosynthesis utilizes H_2S or organics instead of water as electron donors, releasing no oxygen.
- **IAE110100186:** Psychrophiles exhibit optimal metabolic and growth rates at temperatures near or below freezing.
- **IAE110100187:** Endospores contain high concentrations of calcium dipicolinate to stabilize DNA and proteins.
- **IAE110100188:** S-layers are paracrystalline protein surfaces that function as structural barriers in archaea and bacteria.
- **IAE110100189:** Protoplasts are cells whose cell walls have been completely removed enzymatic treatment.
- **IAE110100190:** Sphaeroplasts are Gram-negative cells retaining residual outer membrane structure following wall removal.
- **IAE110100191:** Bdellovibrio is a predatory bacterium that invades and lyses other Gram-negative host cells.
- **IAE110100192:** Bacterial capsule loss directly results in attenuation of virulence.
- **IAE110100193:** Periplasmic space houses hydrolytic enzymes and binding proteins in Gram-negative bacteria.
- **IAE110100194:** Specialized pili mediate twitching motility across solid surfaces.
- **IAE110100195:** Vancomycin inhibits bacterial cell wall synthesis by binding to D-Ala-D-Ala terminals.
- **IAE110100196:** Chlorosomes function as light-harvesting complexes in green sulfur bacteria.
- **IAE110100197:** Membrane lipid branching enhances thermal tolerance in extremophilic archaea.
- **IAE110100198:** Specialized bacterial gas vesicles provide buoyancy control in aquatic environments.

Chapter 07: Protists & Fungi Diversity

- **IAE110100199:** Protists represent a polyphyletic group of eukaryotic microorganisms that do not fit into plants, animals, or fungi.
- **IAE110100200:** Amoeboid movement relies on pseudopodia formation driven by actin-myosin cytoskeleton rearrangement.
- **IAE110100201:** Ciliates utilize cilia for locomotion and macro/micronuclei for somatic and reproductive functions.
- **IAE110100202:** Dinoflagellates possess two perpendicular flagella and cellulose plates, often causing toxic red tides.
- **IAE110100203:** Diatoms feature silica-based frustule cell walls consisting of two overlapping halves.
- **IAE110100204:** Euglenoids lack rigid cell walls, instead possessing a flexible proteinaceous pellicle.
- **IAE110100205:** Plasmodium species exhibit complex life cycles involving Anopheles mosquito vectors and human hosts.
- **IAE110100206:** Slime molds exist as multinucleate plasmodia or cellular aggregations forming fruiting bodies under nutrient depletion.
- **IAE110100207:** Oomycetes possess cell walls made of cellulose rather than chitin and are flagellated heterotrophs.
- **IAE110100208:** Fungal cell walls are composed predominantly of chitin polymers.

- **IAE110100209:** Fungi exhibit absorptive heterotrophy, secreting extracellular enzymes to digest nutrients prior to ingestion.
- **IAE110100210:** Fungal bodies consist of thread-like hyphae forming an interconnected network called a mycelium.
- **IAE110100211:** Septate hyphae are divided by internal cross-walls, whereas coenocytic hyphae lack septa and are multinucleate.
- **IAE110100212:** Plasmogamy is the fusion of cytoplasm from two parent mycelia, preceding nuclear fusion.
- **IAE110100213:** Karyogamy is the fusion of haploid nuclei within a dikaryotic cell to form a diploid zygote.
- **IAE110100214:** The dikaryotic stage (n+n) occurs when haploid nuclei coexist without immediate fusion.
- **IAE110100215:** Chytrids produce flagellated zoospores, representing an early-diverging fungal lineage.
- **IAE110100216:** Zygomycetes form thick-walled zygosporangia capable of surviving unfavorable environmental conditions.
- **IAE110100217:** Ascomycetes produce sexual spores (ascospores) within sac-like structures called asci.
- **IAE110100218:** Basidiomycetes produce sexual spores (basidiospores) externally on club-shaped structures called basidia.
- **IAE110100219:** Deuteromycetes (imperfect fungi) lack an observed sexual reproduction cycle.
- **IAE110100220:** Lichens represent mutualistic symbioses between a photosynthetic alga or cyanobacterium and a fungus.
- **IAE110100221:** Mycorrhizae are mutualistic associations between fungal hyphae and plant root systems.
- **IAE110100222:** Ectomycorrhizae form extracellular sheaths around roots, whereas endomycorrhizae penetrate root cortical cell walls.
- **IAE110100223:** Fungi act as primary decomposers, recycling organic carbon and mineral nutrients in terrestrial ecosystems.
- **IAE110100224:** Contractile vacuoles regulate osmoregulation by pumping excess water out of freshwater protists.
- **IAE110100225:** Apicomplexans possess specialized apical complexes designed to penetrate host cells.
- **IAE110100226:** Red algae contain phycoerythrin pigments enabling photosynthesis at deep ocean depths.
- **IAE110100227:** Brown algae contain fucoxanthin and form large multicellular structures known as kelps.
- **IAE110100228:** Green algae share photosynthetic pigments (chlorophyll a/b) and cell wall features with land plants.
- **IAE110100229:** Choanoflagellates possess a collared flagellum structure matching ancestral animal sponge cells.
- **IAE110100230:** Foraminiferans produce porous calcium carbonate tests through which pseudopodia extend.
- **IAE110100231:** Radiolarians possess intricate internal skeletons made of silica.
- **IAE110100232:** Haustoria are specialized fungal hyphae that penetrate host plant tissues to absorb nutrients.
- **IAE110100233:** Conidia are asexual fungal spores produced externally at the tips of

specialized hyphae.

- **IAE110100234:** Ascocarps are fruiting bodies housing spore-producing asci in Ascomycota.
- **IAE110100235:** Basidiocarps are complex fruiting structures (e.g., mushrooms) produced by Basidiomycota.
- **IAE110100236:** Ergot alkaloids are fungal toxins produced by Claviceps that affect livestock and humans.
- **IAE110100237:** Aflatoxins are carcinogenic compounds produced by Aspergillus species contaminating stored grains.
- **IAE110100238:** Yeasts are unicellular fungi that reproduce predominantly by budding.
- **IAE110100239:** Arbuscular mycorrhizae form highly branched treelike structures within plant root cells for nutrient transfer.
- **IAE110100240:** Soredia are specialized reproductive units of lichens containing both algal and fungal components.
- **IAE110100241:** Trypanosoma is a flagellated protist that causes African sleeping sickness transmitted by tsetse flies.
- **IAE110100242:** Phytophthora infestans is an oomycete pathogen responsible for the historic potato late blight.
- **IAE110100243:** Microsporidia are unicellular intracellular parasites related to fungi, possessing polar tubes for cell entry.
- **IAE110100244:** Parasexual cycles allow genetic recombination in certain fungi without formal meiosis or syngamy.